VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2013
Higher 1

CANDIDATE NAME

CLASS

BIOLOGY 8875/01

Paper 1 Multiple Choice

27 September 2013

1 hour

Additional Materials: Multiple Choice Answer Booklet

READ THESE INSTRUCTIONS FIRST

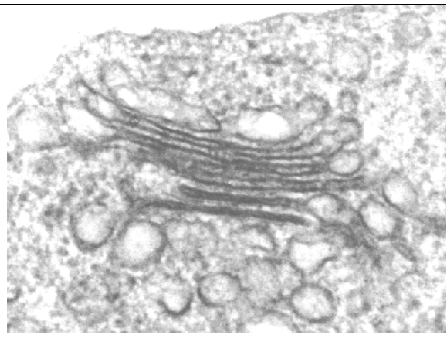
Write in a soft pencil.
Do not use any staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

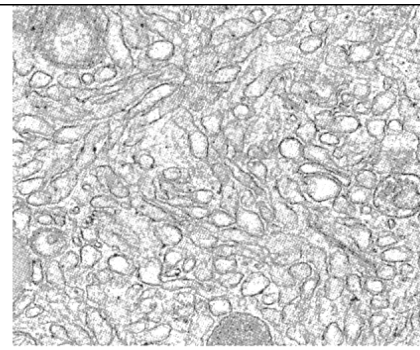
Each correct answer will score one mark. A mark will not be deduced for a wrong answer.
Any rough working should be done in this booklet

1 Below are electron micrographs of four different organelles that can be found in different eukaryotic cells. These micrographs are of different scale.



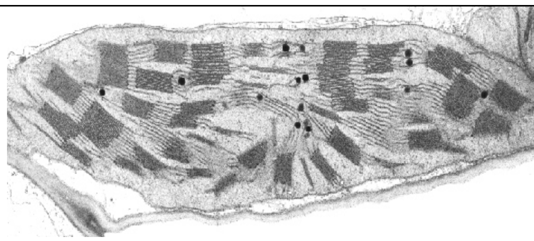
http://biology.unm.edu/ccouncil/Biology_124/Summaries/Cell.html

Organelle M



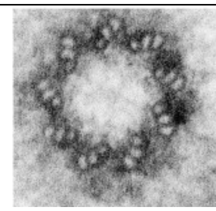
http://wiki.pingry.org/u/ap-biology/index.php/

Organelle N



http://www.biologie.uni-hamburg.de/b-online/e07/07a.htm

Organelle P



http://www.sciencedirect.com/science/article/pii/S0962892410001984

Organelle Q

Which of the following statements is true regarding the above organelles?

	Organelle M	Organelle N	Organelle P	Organelle Q
A	Is the sorting and packaging centre of the cell	Single membrane bound organelle responsible for lipid synthesis	Has a double membrane envelope, circular DNA and 70S ribosomes	Non-membrane bound organelle that is involved in the assembly of spindle fibres
B	Is connected to the rough endoplasmic reticulum so that proteins can be transported to it for packaging	Non-membrane bound organelle that is involved in protein synthesis	Is involved in the synthesis of ATP which is used for cellular activity	Membrane bound organelle involved in the organisation of spindle fibres
C	Double membrane bound organelle involved in modification of proteins	Is involved in the synthesis of carbohydrates	Extensive network of single membrane involved in the synthesis of proteins	Contains rRNA and proteins and are responsible for translation of proteins
D	Single membrane bound organelle responsible for lipid synthesis	Single membrane bound organelle involved in the digestion of food substances	Double membrane organelle containing euchromatin and heterochromatin	Is involved in the synthesis of kinetochore proteins

2 Lysosomal storage diseases are a group of approximately 50 rare inherited metabolic disorders that result from defects in lysosomal function.

Which of the following will be observed in the cells of someone suffering from a lysosomal storage disease?

A The accumulation of monosaccharides due to the lack of enzymes for carbohydrate biosynthesis.

B The accumulation of lipids due to the lack of the enzyme responsible for its hydrolysis.

C The accumulation of salts due to the inability to remove minerals.

D The overproduction of lysosomal membrane by the rough endoplasmic reticulum and Golgi apparatus.

3 A sample of yeast cells were grown in a culture. Radioactive amino acids were added to the solution in which they were being grown. At various times, samples of the cells were taken and the amount of radioactivity in different organelles was measured. The results are shown in the table.

Time after radioactive amino acids were added to the solution/ minutes	Amount of radioactivity present/arbitrary units		
	P	**Q**	**R**
1	21	120	6
20	42	68	6
40	86	39	8
60	76	28	15
90	50	27	28
120	38	26	56

Which best describes the identities of organelles P, Q and R?

	P	**Q**	**R**
A	Golgi apparatus	Rough endoplasmic reticulum	Secretory vesicles
B	Rough endoplasmic reticulum	Smooth endoplasmic reticulum	Golgi apparatus
C	Rough endoplasmic reticulum	Nucleus	Golgi apparatus
D	Smooth endoplasmic reticulum	Golgi apparatus	Secretory vesicles

4 **Figure 1** below shows the structure of a biomolecule extracted from a cell. It is made up of two components **S** and **T**. **Figure 1a** shows a magnified portion of the **component S**. **Figure 1b** refers to the other component, **component T**, of the same molecule.

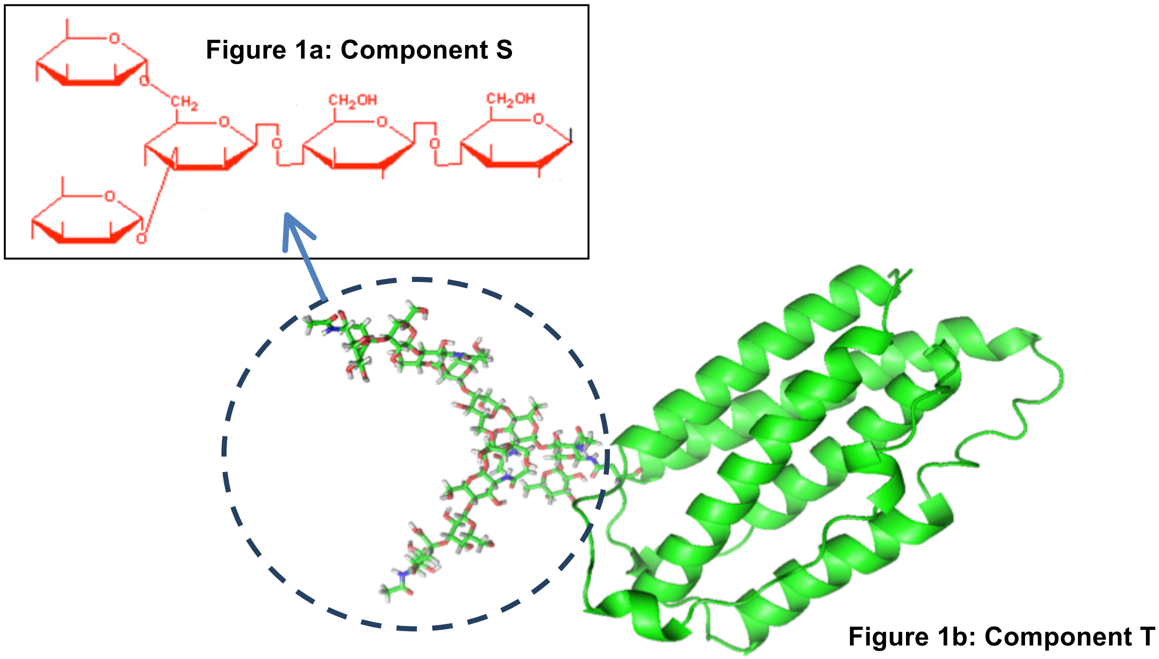


Figure 1: Structure of biomolecule

Below are some statements regarding the structure, property and function of biomolecules with structure shown in **Figure 1**. Which of the statements are true?

(i) The biomolecule has both hydrophilic and hydrophobic properties.

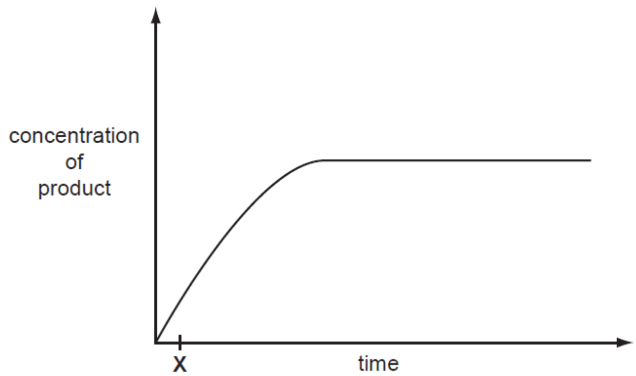
(ii) It is important for cell to cell adhesion

(iii) The synthesis of this molecule requires only the rough endoplasmic reticulum.

(iv) When completely hydrolysed, all the monomers of this biomolecule are soluble in water.

A **(i)** and **(iii)**
B **(ii)** and **(iv)** only
C **(i)**, **(ii)** and **(iv)** only
D **(ii)**, **(iii)** and **(iv)** only

5 The graph shows the course of an enzyme-catalyzed reaction at 30°C.



What is true at time **X**?

A Most enzyme molecules will have free active sites.

B The number of available substrate molecules is high.

C The number of enzyme-substrate complexes is low.

D The rate remains the same if more enzyme is added.

6 Which of the following statements is **false** of collagen and haemoglobin?

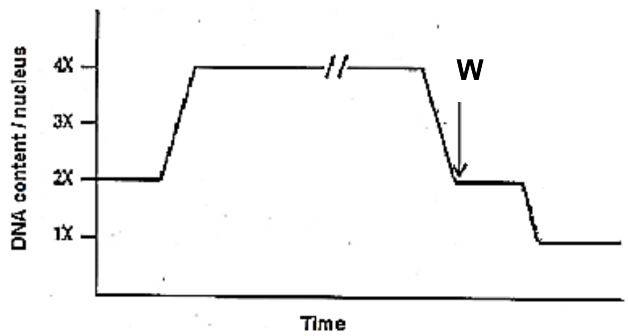
A Both are made up of amino acids.

B Both exhibit quaternary level of organisation.

C Collagen has a branched structure whereas haemoglobin has a globular structure.

D There is less variability in the monomers present in collagen compared to haemoglobin.

7 During metaphase I in a plant cell, six bivalents were aligned along the equator of the cell. The graph below shows the changes in the amount of DNA in various stages in meiosis of the same plant cell.



Based on the information provided, which of the following shows correctly the number of chromosomes and DNA molecules for a cell in stage **W**?

	Number of chromosomes	Number of DNA molecules
A	6	12
B	12	12
C	6	24
D	12	24

8 Male bees are haploid. They develop from unfertilised eggs. Female bees are diploid.

Which of the following statements are true?

1 All male bees are genetically identical.

2 Male bee sperm cells are produced by mitosis.

3 New combinations of genes only occur in female bees.

A 1 and 2 only

B 1 and 3 only

C 2 and 3 only

D 1, 2 and 3

9 *Staphylococcus aureus*, a bacterium, has a genome size of 2.8×10^6 base pairs. During DNA replication, 400 base pairs are formed per second at one replication fork.

How long will it take to replicate the entire *Staphylococcus aureus* genome?

A 29 minutes

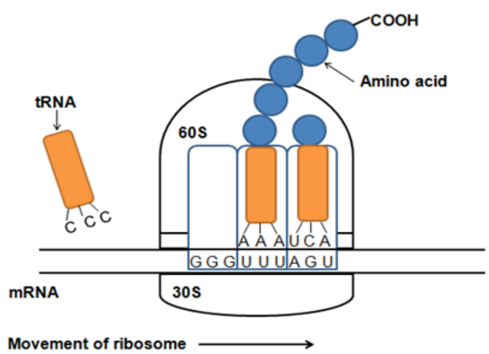
B 58 minutes

C 116 minutes

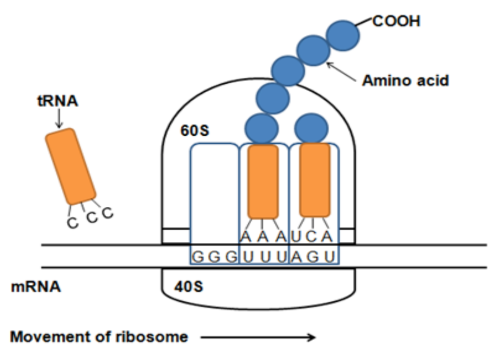
D 233 minutes

10 Which of the following diagrams show the correct process of eukaryotic translation?

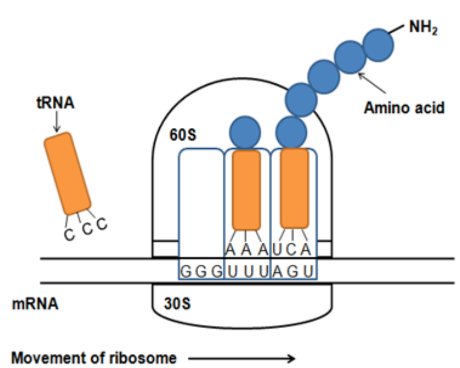
A



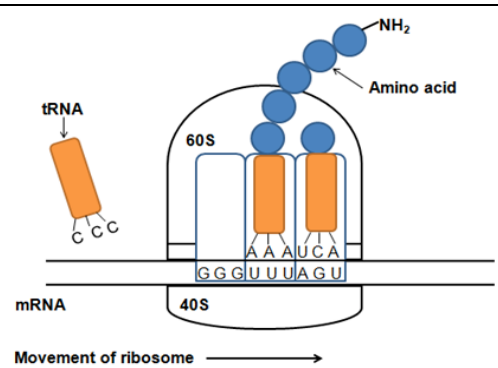
B



C



D



11 Fabry's Disease involves a mutation that occurs in the gene that codes for α-galactosidase A.

Normal allele

Codon	37	38	39	40	41	42	43	44	45
mRNA	CCU	UGG	ACC	CAG	AGG	UUC	UAA	GAC	GGA

Mutated allele

Codon	37	38	39	40	41	42	43	44	45
mRNA	CCU	UGG	ACC	CCG	CAG	AGG	UUC	UAA	GAC

Which of the following statement(s) is true about the type and effect of this mutation?

(i) It is a frame-shift mutation

(ii) It involves an insertional mutation

(iii) It will result in the lengthening of polypeptide chain

(iv) It is a chromosomal duplication

A **(i)** only

B **(i)** and **(ii)** only

C **(ii)** and **(iii)** only

D **(i), (iii)** and **(iv)**

12 In guinea pigs, the allele for rough coat is dominant over the allele for smooth coat and the allele for black fur is dominant over the allele for white fur. The genes for fur colour and texture are on 2 different chromosomes

Two guinea pigs which are heterozygous for both fur colour and texture were mated together. What is the probability that the offspring would be homozygous for one of the dominant alleles?

A 1 in 2

B 1 in 3

C 3 in 8

D 2 in 9

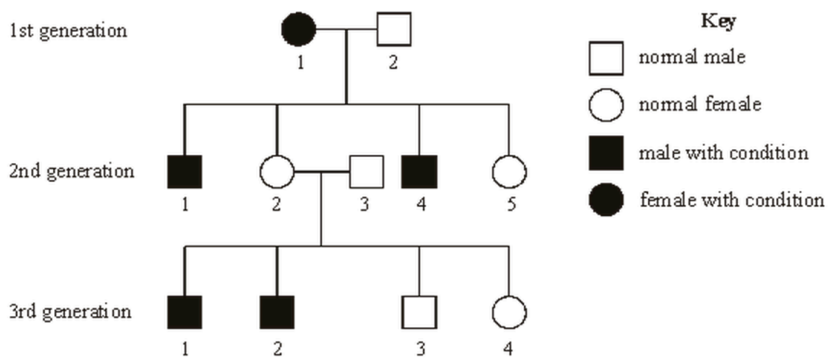
13 Coat colour in rabbits is controlled by a gene with four alleles. The order of dominance for these alleles is as follows:

agouti (**C**) > chinchilla ($\mathbf{C^c}$) > himalayan ($\mathbf{C^h}$) > albino (**c**)

What is the maximum number of classes of coat colour obtained from a cross between an agouti rabbit and a himalayan rabbit?

A 1 **B** 2 **C** 3 **D** 4

14 The diagram below shows the pedigree of a family with a disease condition.



If individual 4 from the 3rd generation marries a normal male, what is the probability that the first offspring is a normal male?

A 1/4 **B** **1/3** **C** 1/2 **D** 3/4

15 In an experiment, six tubes were set up as shown in the table below.

tube	contents
1	Glucose + homogenized animal cells
2	Glucose + mitochondria
3	Glucose + cytoplasm lacking organelles
4	Pyruvate + homogenized animal cells

Provided all other conditions are kept constant, which of the following shows the amount of ATP produced in each tube in **increasing** order?

A 1 – 3 – 4 – 2

B 3 – 2 – 1 – 4

C 2 – 3 – 4 – 1

D 4 – 2 – 3 – 1

16 From yeast cells, a mixture of enzymes can be obtained that catalyses the conversion of glucose to carbon dioxide and ethanol.

Which combination of substances would increase the rate of ethanol production by these enzymes?

A ATP and lactate

B Reduced NAD and coenzyme A

C ATP and reduced NAD

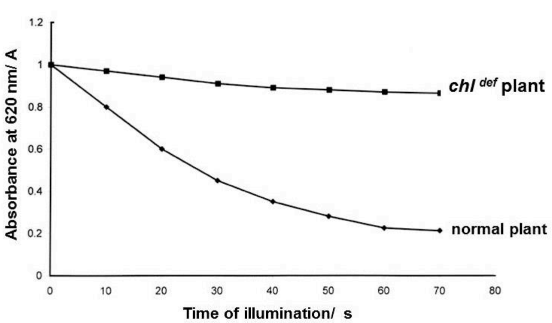
D NAD and carbon dioxide

17 Dichlorophenol-indophenol (DCPIP) is used to investigate the light-dependent stage of photosynthesis. DCPIP is a blue dye which acts as an electron acceptor. As DCPIP is reduced, its blue colour disappears. The depletion of DCPIP is indicated by a drop in absorbance.

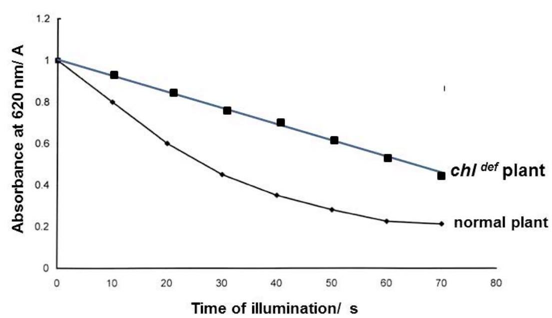
Leaf extracts were prepared from plants with normal *chlorophyll a* gene and plant that contain a deletion mutation in *chlorophyll b* gene (*chl*^{def}), thus reducing the amount of chlorophyll b present. These extracts were mixed with DCPIP solution and exposed to white light for different lengths of time and absorbance were determined.

Identify the correct set of results.

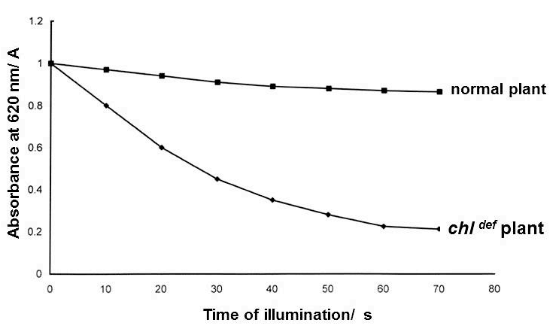
A



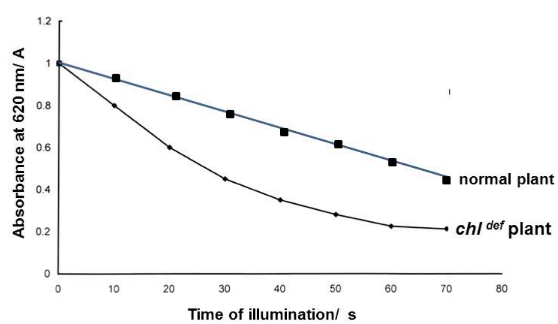
B



C

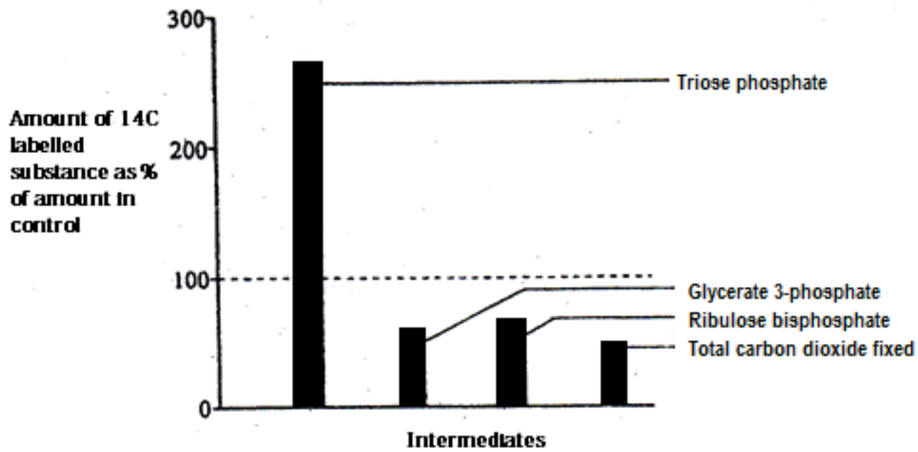


D



18 An experiment was conducted to test the properties of a chemical G on the photosynthetic capabilities of a unicellular agar, *Chlorella*. An illuminated suspension of the alga was treated with carbon dioxide labeled with ^{14}C in the presence of an unknown chemical G. The light was switched off and the amount of radioactivity present in some intermediates was determined after 10 minutes in the dark.

A control suspension of alga without chemical G being added was treated in exactly the same manner. The bar chart below shows the amount of radioactivity in these intermediates in the alga with chemical G added as a percentage of the intermediates in the control alga.



Which option correctly describes the action of chemical G?

A G binds to NADPH produced in light reactions and prevent its oxidation process.

B G competes with triose phosphate for the active site of the enzyme that converts triose phosphate into hexose phosphate.

C G prevents the regeneration of ribulose bisphosphate at the stage after triose phosphate was formed.

D G inhibits the ribulose bisphosphate carboxylase enzyme, preventing carbon fixation from taking place efficiently.

19 Which of the following is **not** a valid comparison between oxidative phosphorylation and photophosphorylation?

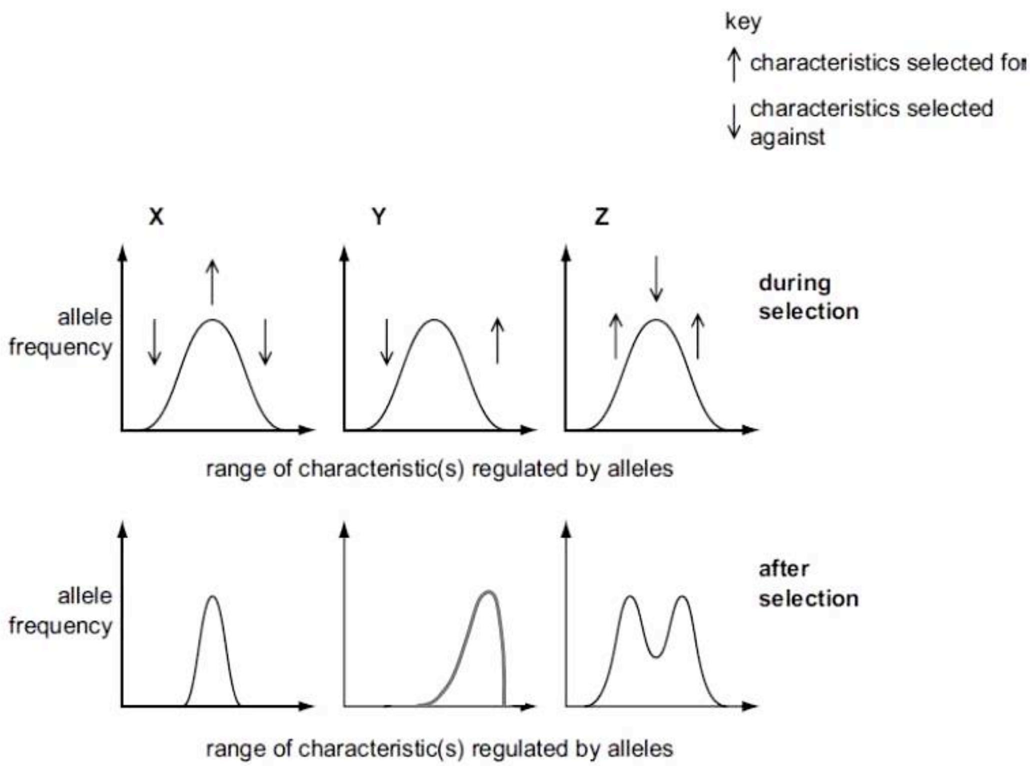
A Both utilize energy from the movement of electrons along the electron transport chain to accumulate protons in a given space.

B Solar energy is the main energy source for photophosphorylation while stored chemical energy is the main source of energy for oxidative phosphorylation.

C Both oxidative phosphorylation and photophosphorylation are important for channelling proton motive force into the synthesis of ATP.

D Water acts as the final electron acceptor in oxidative phosphorylation while it acts as the electron donor in photophosphorylation.

20 The graphs represent the frequency of alleles in species X, Y and Z during and after selection.



In which species has evolution taken place after selection?

A X only

B Y only

C Y and Z only

D None of the above

21 Which of the following is not an evidence for evolution?

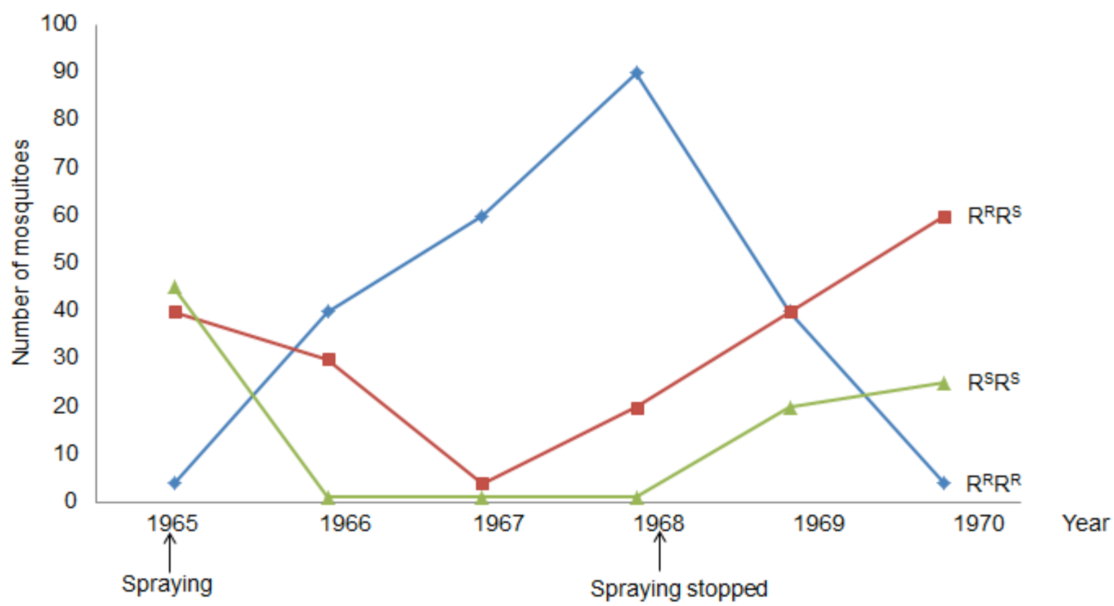
A A cactus spine and a carrot are modified part of a plant

B Wingless bird, kiwi, has the presence of wing bones

C Presence of gills slits in human embryos.

D Codon coding for a particular amino acid

22 In the mosquito there is a gene locus which has two alleles, R^R = Resistant and R^S = Sensitive, which are involved in resistance, in particular to the insecticide DDT.
The graphs below show the number of mosquitoes of the 3 phenotypes (and genotypes) collected from 1965, when DDT was first used, through 1970, two years after the spraying of DDT stopped



From the data it is possible to conclude that

A the frequency of the R^S allele is greater than the frequency of the R^R allele in 1968.

B many generations after the removal of DDT the R^R allele would disappear from the population.

C after the removal of DDT from the environment in 1968, having the R^RR^R genotype reduces the chance of survival.

D between 1967 and 1968 in the presence of DDT in the environment, the mosquitoes with the R^RR^S genotype are the most likely to survive.

23 *Anolis sagrei*, a lizard native to Bahamas Island, spend most of their time on ground. A predatory lizard, *Leiocephalus carinatus*, a ground dwelling predator of Anolis sagrei was introduced.

6 months later, most of the *Anolis sagrei* lizards found in the island have longer legs.
12 months later, most of the *Anolis sagrei* lizards found in the island have shorter legs but becoming increasingly arboreal, spending time in treetop.

Which of the following statement(s) is/are true

(i) Presence of the predatory lizard results in mutation *of Anolis sagrei* gene coding for the length of leg

(ii) *Anolis sagrei* with longer legs can run faster and hence are at an advantage initially

(iii) *Anolis sagrei* being arboreal can better evade *Leiocephalus carinatus.*

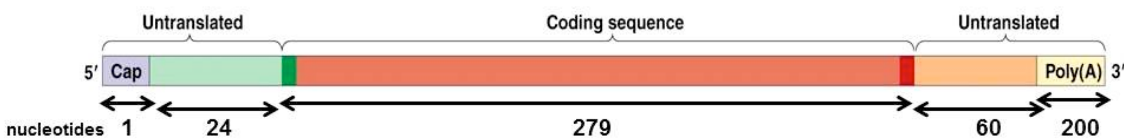
A **(i)**

B **(i)** and **(ii)**

C **(ii)** and **(iii)**

D None of the above

24 The mature eukaryotic mRNA contains the structure shown below.



When the reverse transcription of this mRNA is performed for insertion into a bacterial plasmid for transforming E coli cells, the length of the resulting cDNA fragment will be

A 279 bp
B 303 bp
C 363 bp
D 564 bp

25 Familial hypercholesterolaemia (FH) increases risk of premature coronary artery disease.

Fig. 2a shows the pedigree of a family with members **A** to **H** on whom DNA analysis was performed (**Fig. 2b**). Individual **H** has high cholesterol levels due to the inheritance of a mutant allele of the *LDL* receptor gene.

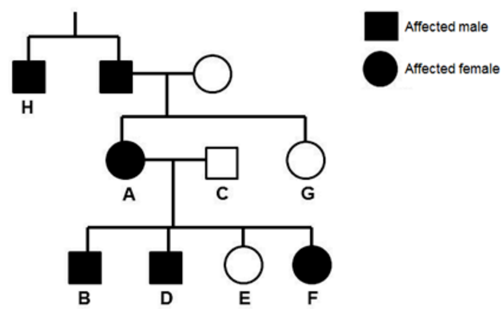


Fig. 2a

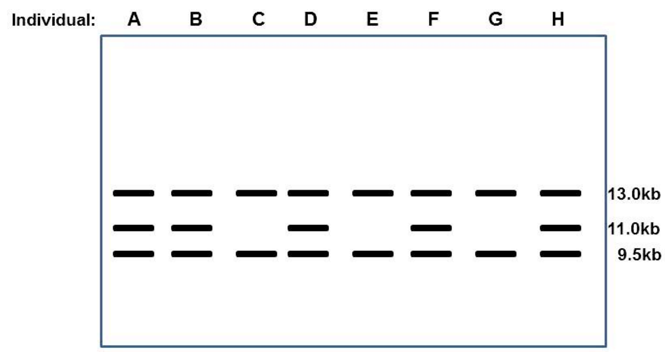


Fig. 2b

Based on **Fig. 2a** and **Fig. 2b**, what conclusions can be correctly drawn?

A	FH is inherited in an autosomal dominant manner.	The 11.0kb band is the result of the mutant allele.
B	FH is inherited in an autosomal dominant manner.	The 13.0kb band is the result of the mutant allele.
C	FH is inherited in a sex-linked recessive manner.	The 13.0kb and 9.5kb bands are the result of the mutant allele.
D	FH is inherited in a sex-linked recessive manner.	The 11.0kb band is the result of the mutant allele.

26 Which of the following DNA sequences would a restriction enzyme usually recognize and cut?

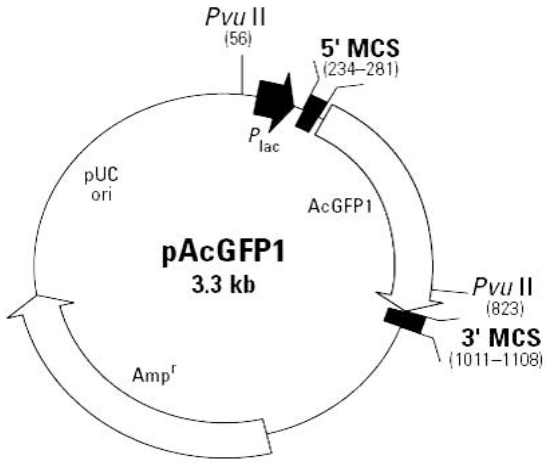
A 5' – ATGCAC – 3'
3' – TACGTG – 5'

B 5' – GATATC – 3'
3' – CTATAG – 5'

C 5' – TAGATA – 3'
3' – ATCTAT – 5'

D 5' – AATATA – 3'
3' – TTATAT – 5'

27 Plasmid vectors are used in the genetic engineering of desirable genes for introduction into bacteria in the production of human insulin and growth hormones for use in treating insulin and growth hormone deficiency. A plasmid vector is shown below.



Identify the functions of the following features of a plasmid vector.

	Origin of replication	Selectable marker	Multiple cloning site
A	Replicate simultaneously with host genome	Can be induced at desired time when required	Contains many enzyme restriction sites for the insertion of foreign genes from different species
B	Allows recombinant plasmids to replicate independently from host DNA and generate numerous copies of the inserted gene	Codes for identifiable phenotype to help the identification of cells that contains the inserted gene	Contains many enzyme restriction sites for the insertion of foreign genes from different species
C	Replicate simultaneously with host genome	Codes for identifiable phenotype to help the identification of cells that contains the inserted gene	Digest viral DNA into fragments
D	Allows recombinant plasmids to replicate independently from host DNA and generate numerous copies of the inserted gene	Can be induced at desired time when required	Digest viral DNA into fragments

28 What is the purpose of the Human Genome Project?

A To clone every gene on a single chromosome in human DNA
B To splice every gene on a single chromosome in human DNA
C To inbreed the best genes on every chromosome in human DNA
D To identify the DNA sequence of every gene in the human genome

29 Which of the following statements is/are true regarding zygotic stem cells and cancer cells

(i) Both are able to move one from location to another

(ii) Both are able to divide by mitotic division

(iii) Both are specialized cells and capable to differentiate further

(iv) Both are capable of indefinite replication

A **(ii)** only
B **(i)** and **(iii)** only
C **(ii)** and **(iv)** only
D **(i), (ii)** and **(iv)** only

30 Which of the following is not a valid concern about the use of Bt corns

A Affecting non-target insects
B The decreased usage of herbicide
C Pollens of the Bt corns are wind borne
D Possible allergic reaction when consumed by human